arithmetic mistakes, and more. For each dataset, we explain how it covers something different from all the previous datasets. For all datasets, we present the LLM with non or a short instruct, a context (if exists for the task), and let it generate a free text. We follow this paradigm as it better mimics real-world usage of LLMs by humans, as opposed to using few-shot to force a short answer that is generated on the first token (Yuksekgonul et al., 2023; Chen et al., 2024; Simhi et al., 2024). One exception to this is a the sentiment analysis (IMDB) for which we apply 1-shot for the LLM to use the allowed labels, as it did not follow the instruction alone and we could not identify if the answer is correct or not even with manual analysis. Additionally, we implemented a different prompting strategy to the instruct and non-instruct LLMs. To see the exact formats we used to prompt each dataset and LLM, refer to our code implementation at `https://github.com/technion-cs-nlp/LLMsKnow`.

For each dataset we used a split of 10K training samples and 10K test samples, unless the dataset is too small, in which case we mention the size.

- **TriviaQA** (Joshi et al., 2017): a collection of trivia question-answer pairs. The questions are presented to the LLM without any context, allowing it to generate responses based solely on its internal, parametric knowledge. The dataset includes various acceptable variations of the correct answer, which are used to automatically evaluate the accuracy of the generated res.
- **HotpotQA** (Yang et al., 2018): a dataset designed for diverse multi-hop question answering. Each entry includes Wikipedia documents that help answering the questions. We use two different settings: (1) without context, where questions are asked directly, which covers slightly different skills from TriviaQA as it requires reasoning in addition to factual knowledge; and (2) with context (**HotpotQA_WC**), where the additional context is provided, emphasizing the ability to adhere to and utilize contextual information to solve the task.
- **Movies:** to further investigate generalization, we focused on a case of classic "hallucinations", involving factual knowledge, within a non-diverse dataset. This approach allowed us to test whether generalization to other types of errors is influenced by the type of error (factual versus others) or by the dataset's diversity. For this purpose, we created the movies dataset consisting of prompts in the form: "Who acted as [figure name] in the movie [movie name]?" The figures, movies, and correct answers were sourced from "The Movies Dataset" in Kaggle: `https://www.kaggle.com/datasets/rounakbanik/the-movies-dataset`, which is based on the MovieLens website.
- **Winogrande** (Sakaguchi et al., 2021): we use this dataset to explore errors in common-sense reasoning. It consists of Winograd-style coreference challenges, where each example presents a sentence containing two entities and a pronoun. The objective is to determine which entity the pronoun refers to, relying on common-sense reasoning. For example, in the sentence: "The trophy doesn't fit into the suitcase because it's too **large**," the pronoun "it" refers to the trophy, not the suitcase.
- **Winobias** (Zhao et al., 2018): this benchmark focuses on coreference resolution in the context of gender bias, revealing a different type of limitation in LLMs. Each example consists of two professions: one stereotypically male and one stereotypically female, along with a gendered pronoun. The task requires the LLM to determine which profession the pronoun refers to. The sentences are unambiguous, with one correct answer. In some cases, the correct answer aligns with the stereotype, while in others, it is anti-stereotypical. For example, in the sentence "The developer argued with the designer because she did not like the design," "she" refers to the developer, which is an anti-stereotypical case since "developer" is considered a stereotypically male profession. Research has shown that LLMs often perform poorly on anti-stereotypical sentences (Zhao et al., 2018) and tend to base their decisions on stereotypes rather than on common-sense reasoning or linguistic rules (Kotek et al., 2023). Each split contains around 1500 samples.
- **NLI** (Natural Language Inference): NLI involves determining whether a given "hypothesis" is true (entailment), false (contradiction), or undetermined (neutral) based on a provided "premise." For this purpose, we use the MNLI dataset (Williams et al., 2018). NLI tasks address a distinct aspect of common-sense reasoning and are generally considered complex. This complexity allows us to investigate whether a model's generalization ability

is related to the difficulty of the task it was trained on, or to other factors, such as the limited diversity of labels (NLI has only three valid labels) or the type of task.

- **Math** (Sun et al., 2024): this dataset includes both unanswerable and answerable math problems. In our study, we focus exclusively on the answerable problems, as our aim is to assess the correctness of the LLM's outputs, which requires a known correct answer (gold standard). This task introduces an additional, previously unexplored skill of arithmetic reasoning. The train-test split consists of approximately 2,000 and 650 samples, respectively.
- **IMDB** (Maas et al., 2011): contains movie reviews used for the task of sentiment classification.
- **Natural Questions With Context** (Kwiatkowski et al., 2019): the Natural Questions (NQ) dataset is designed to evaluate and train automatic question-answering systems. It consists of real, anonymized queries submitted by users to Google, with answers extracted from Wikipedia, as well as the relevant Wikipedia pages which can be given in context. We included this dataset to introduce an additional challenge that requires adherence to context, complementing the HotpotQA with context dataset.

A.4 Baselines: Implementation Details

Aggregated probabilities / logits. Inspired by prior work (Kadavath et al., 2022; Guerreiro et al., 2023), we compute an aggregated score using the log-probabilities or raw probabilities of the generated text tokens $y_1, y_2, \ldots, y_N$ produced by the generative large language model (LLM). For instance, the following formulation is used to compute the Logits-mean baseline on the entire generated answer:

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{P}(y_i|Q, y_1, ..., y_{i-1}) \tag{1}$$

We also explore aggregation strategies that focus solely on the exact answer tokens (PE-Exact). Following Varshney et al. (2023), we also experiment with aggregating the minimum and maximum values (PE-[Min—Max]-[Exact]), alongside the mean aggregation described in Equation 1.

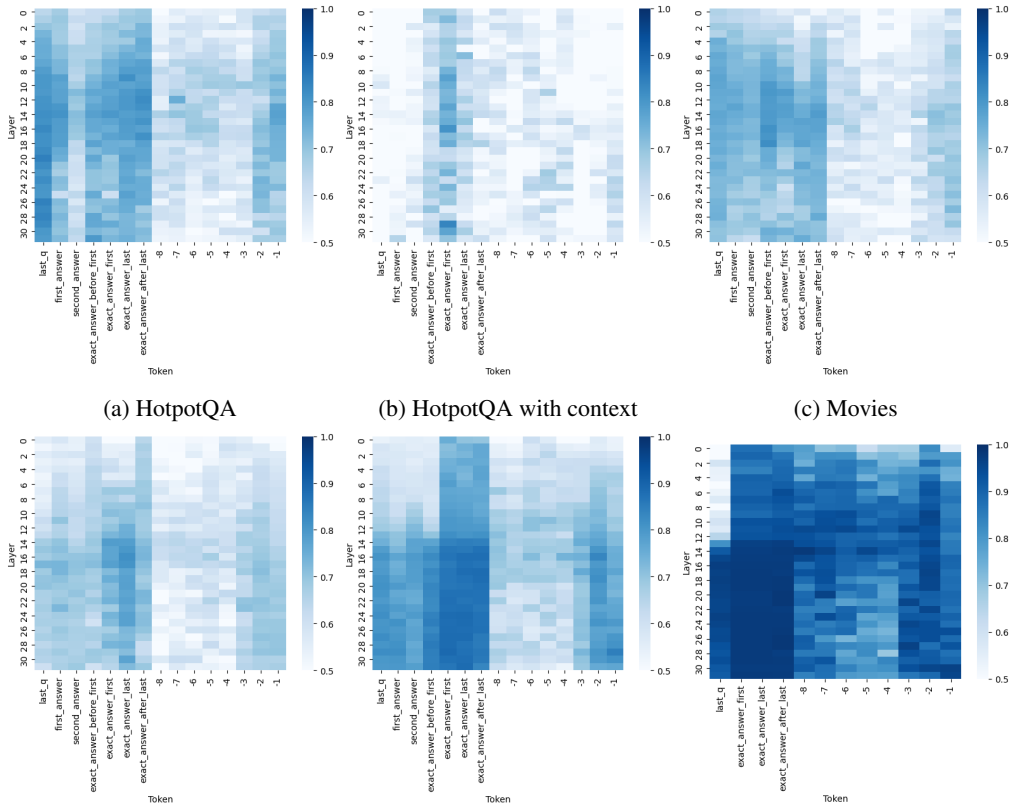
P(True): We follow Kadavath et al. (2022) and prompt the LLM to judge whether its answer is correct. Our prompt followed the following template, from Kadavath et al. (2022):

> Question: `[Question]`
>
> Proposed Answer: `[LLM long answer]`
>
> Is the proposed answer:
>
> (A) True
>
> (B) False
>
> The proposed answer is:

B Full Error Detection Results

Figure 6 presents the AUC values of a traind probe across layers and token for Mistral-7b-instruct, showing a similar pattern across all datasets. We also observe similar patterns across other models. See our repo `https://github.com/technion-cs-nlp/LLMsKnow`for the figures.

Tables 4, 5, 6, and 7 present the full error detection results across all baselines and datasets, which are consistent with the main paper results.



(a) HotpotQA (b) HotpotQA with context (c) Movies

(d) Winogrande (e) NLI (f) IMDB

Figure 6: AUC values of a probe error detector across layers and tokens, Mistral-7b-instruct. The detection performance spikes at the exact answer tokens.

Table 4: Comparison of error detection performance (AUC) on Mistral-7B.

	Mistral-7B				
	TriviaQA	Winobias	Math	Movies	IMDB
Logits-mean	0.67 ±0.004	0.49 ±0.010	0.41 ±0.015	0.67 ±0.007	0.88 ±0.064
Logits-mean-exact	0.67 ±0.004	0.50 ±0.010	0.56 ±0.026	0.68 ±0.008	0.57 ±0.080
Logits-min	0.80 ±0.003	0.45 ±0.014	0.48 ±0.021	0.73 ±0.006	0.78 ±0.056
Logits-min-exact	0.80 ±0.005	0.53 ±0.014	0.78 ±0.032	0.72 ±0.005	0.57 ±0.080
Logits-max	0.53 ±0.008	0.49 ±0.010	0.42 ±0.023	0.54 ±0.005	0.83 ±0.076
Logits-max-exact	0.54 ±0.009	0.50 ±0.010	0.40 ±0.024	0.58 ±0.007	0.57 ±0.080
Probas-mean	0.76 ±0.003	0.53 ±0.018	0.66 ±0.016	0.72 ±0.007	0.87 ±0.041
Probas-mean-exact	0.78 ±0.002	0.55 ±0.014	0.62 ±0.016	0.74 ±0.007	0.83 ±0.057
Probas-min	0.82 ±0.003	0.52 ±0.013	0.82 ±0.020	0.73 ±0.006	0.86 ±0.032
Probas-min-exact	**0.85** ±0.003	0.58 ±0.011	0.84 ±0.015	0.74 ±0.006	0.83 ±0.057
Probas-max	0.53 ±0.008	0.50 ±0.016	0.43 ±0.025	0.55 ±0.008	0.80 ±0.074
Probas-max-exact	0.55 ±0.009	0.51 ±0.013	0.39 ±0.019	0.59 ±0.009	0.83 ±0.057
p(True)	0.57 ±0.007	0.53 ±0.019	0.56 ±0.027	0.51 ±0.003	0.65 ±0.004
p(True)-exact	0.56 ±0.006	0.55 ±0.026	0.57 ±0.036	0.52 ±0.003	0.65 ±0.003
Probe @ token					
Last generated [-1]	0.83 ±0.002	0.65 ±0.008	0.82 ±0.023	0.79 ±0.002	0.85 ±0.007
Before last generated [-2]	0.82 ±0.003	0.84 ±0.012	0.83 ±0.019	0.78 ±0.003	0.95 ±0.004
End of question	0.74 ±0.005	0.78 ±0.012	0.83 ±0.016	0.77 ±0.002	0.81 ±0.009
Exact answer last	0.84 ±0.005	**0.89** ±0.007	**0.96** ±0.008	0.78 ±0.003	**0.95** ±0.004
Exact answer last+1	0.84 ±0.004	0.84 ±0.012	0.95 ±0.010	**0.80** ±0.002	0.85 ±0.007
	HotpotQA	HotpotQA-WC	Winogrande	NLI	NQ-WC
Logits-mean	0.63 ±0.005	0.52 ±0.009	0.49 ±0.004	0.51 ±0.004	0.69 ±0.006
Logits-mean-exact	0.57 ±0.008	0.52 ±0.007	0.50 ±0.003	**0.93** ±0.004	0.72 ±0.005
Logits-min	0.72 ±0.008	0.59 ±0.006	0.50 ±0.007	0.53 ±0.005	0.65 ±0.009
Logits-min-exact	0.72 ±0.007	0.65 ±0.004	0.51 ±0.007	0.49 ±0.006	0.70 ±0.005
Logits-max	0.54 ±0.007	0.49 ±0.010	0.48 ±0.005	0.48 ±0.005	0.59 ±0.012
Logits-max-exact	0.48 ±0.010	0.44 ±0.007	0.50 ±0.003	0.48 ±0.005	0.58 ±0.009
Probas-mean	0.65 ±0.004	0.55 ±0.006	0.51 ±0.007	0.49 ±0.003	0.63 ±0.008
Probas-mean-exact	0.62 ±0.006	0.56 ±0.007	0.51 ±0.005	0.02 ±0.001	0.66 ±0.007
Probas-min	0.73 ±0.005	0.58 ±0.007	0.52 ±0.009	0.53 ±0.004	0.63 ±0.011
Probas-min-exact	0.78 ±0.005	0.66 ±0.004	0.52 ±0.008	0.49 ±0.005	0.69 ±0.006
Probas-max	0.54 ±0.008	0.49 ±0.007	0.50 ±0.005	0.47 ±0.004	0.52 ±0.004
Probas-max-exact	0.48 ±0.010	0.44 ±0.005	0.50 ±0.004	0.48 ±0.003	0.53 ±0.012
p(True)	0.55 ±0.007	0.54 ±0.006	0.51 ±0.005	0.51 ±0.003	0.52 ±0.008
p(True)-exact	0.61 ±0.005	0.54 ±0.006	0.61 ±0.006	0.51 ±0.006	0.53 ±0.014
Probe @ token					
Last generated [-1]	0.78 ±0.006	0.67 ±0.004	0.51 ±0.007	0.77 ±0.004	0.78 ±0.003
Before last generated [-2]	0.79 ±0.007	0.69 ±0.007	0.66 ±0.004	0.81 ±0.002	0.75 ±0.006
End of question	0.72 ±0.007	0.56 ±0.003	0.51 ±0.007	0.88 ±0.004	0.70 ±0.005
Exact answer last	0.80 ±0.008	**0.74** ±0.007	**0.69** ±0.006	0.84 ±0.004	0.81 ±0.009
Exact answer last+1	**0.81** ±0.008	0.72 ±0.005	0.59 ±0.005	0.75 ±0.006	**0.84** ±0.007